
Before the Chokepoint: Hyperscaler Embeddedness in Emerging Quantum Network Infrastructure

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Abstract

Dominant firms are usually understood to entrench market power through prices, contracts and acquisitions. This paper argues that in emerging technology domains that power is also reproduced upstream, in the organisation of knowledge production, where the dependencies that later appear as infrastructural lock-ins are first seeded and developed. We examine quantum networking, a pre-material infrastructure whose most fully realised applications, i.e. blind quantum computing and entanglement-based networking, would render the centralised intermediation on which hyperscale cloud providers depend technically unnecessary. These same providers are among the field's most active sponsors.

The paper introduces a new firm-level dataset of quantum internet-enabling research by seven major hyperscalers (Amazon, Microsoft, Google, IBM, Alibaba, Baidu and Oracle), constructed from the OpenAlex scholarly graph and covering approximately 10,000 publications from 2015 to 2024. Combining descriptive network analysis with exponential random graph modelling, we examine where hyperscalers embed in national research ecosystems, which layers of the quantum stack they prioritise, and whether observed co-production patterns reflect strategic positioning beyond what national research productivity and prior collaboration history alone predict.

We find that embeddedness is geographically concentrated, topically skewed toward computing-oriented stack layers at the expense of decentralisation-enabling networking research, and partially structured by firm nationality net of baseline network tendencies. National quantum programme status does not significantly predict hyperscaler tie formation, suggesting that state-led sovereignty initiatives have not, in the short run, redirected corporate research networks. This is a finding that bears directly on global debates over technological sovereignty, economic security, and the role of private authority in exercising infrastructural power. We identify the alternative explanations the current data cannot resolve.

1 Introduction

Cloud computing has become a core enabler of state capacity. With a growing number of government departments and public institutions relying on private-sector cloud services for data storage and service delivery, a small number of dominant firms, known as hyperscalers, increasingly mediate where data is stored, where computation occurs, and under which jurisdictions those processes can be governed. Hyperscalers are distinguished from ordinary cloud providers by the scale at which they operate: millions of physical and virtual servers across globally distributed data centres, delivering high-performance computing, storage, and networking as on-demand services. For governments, particularly in Europe, the cloud has consequently moved from being a smart procurement choice for offloading data storage to mission-critical infrastructure that has introduced significant structural dependencies.

International political economy has developed powerful vocabularies for making sense of the global rise of hyperscalers. Research on the changing character of private authority, the governance of markets, and platform ecosystems has produced a wealth of insight regarding the international security implications of asymmetric infrastructural control in economies with significant network effects ([Bomont, 2025](#); [Calderaro and Marzouki, 2022](#); [DeNardis et al., 2020](#); [Musiani et al., 2015](#); [Rochet and Tirole, 2003](#); [Ward et al., 2011](#)). One dimension, however, remains empirically under-specified: how such power is extended into *future* infrastructures. This paper asks, to what extent, and through what structural features, are hyperscalers embedded in the transnational knowledge networks that are constituting future networking technologies as an emerging infrastructure?

This question matters greatly as market dominance is not reproduced only through prices, contracts, or acquisitions but increasingly reproduced upstream, this paper argues, through the organisation of knowledge production itself: hyperscalers aim to shape which technical problems attract resources, which research communities become legible to investment, and which interfaces define what is interoperable and governable. This upstream dimension matters because it is where future dependencies are seeded before they crystallise later as visible lock-in.

This article examines quantum networking as a new site of such upstream positioning. Quantum internet-enabling technologies are frequently framed as security-enhancing and, in some applications, privacy-preserving. In its strongest form, blind quantum computing promises computation on remote servers without revealing

inputs, processes, or outputs—a conceptual challenge to data-hungry business models. Yet the ecosystem developing the quantum stack does not resemble a decentralising, privacy-first counter-movement. It increasingly features the same actors that consolidated control over the classical internet’s service layer: hyperscale cloud providers, national laboratories, and security-adjacent research communities.

To make this upstream politics observable, the paper shifts attention from deployments to knowledge production. At this stage, quantum networking is a pre-material infrastructure: experimental testbeds, standards conversations, and engineering roadmaps exist, but large-scale operational networks do not. In such contexts, research ecosystems are not peripheral but the central venues where agendas crystallise and where future control points can be positioned. Who collaborates with whom, which parts of the stack are prioritised, and which funders channel resources into particular lines of work are tractable empirical signals of how infrastructural power is assembled upstream.

Empirically, the paper introduces a new dataset constructed from the [OpenAlex](#) scholarly graph, covering quantum internet-enabling research produced by seven major hyperscalers (Amazon, Microsoft, Google, IBM, Alibaba, Baidu, and Oracle) across approximately 10,000 publications from 2015 to 2024. The dataset is enriched with a hyperscaler attribution layer and harmonised institutional and country affiliations. To our knowledge, no existing bibliometric study of quantum technology uses the firm as its primary unit of analysis: prior work has mapped national and institutional patterns ([Coccia, 2024](#); [Conde-Torres et al., 2026](#); [Jiang and Chen, 2021](#); [Scheidsteger et al., 2022](#); [Wang et al., 2021](#)), but the distinctive position of individual corporate actors within those landscapes has not been systematically examined.

The paper asks: *to what extent, and through what structural features, are hyperscalers embedded in the transnational knowledge networks that are constituting quantum networking as an emerging infrastructure?* To answer this, we combine descriptive network statistics with an exponential random graph model (ERGM), which tests whether observed co-production tie patterns reflect hyperscaler-specific attributes and geographic covariates rather than network-endogenous processes alone, thus separating genuine embeddedness signals from the baseline tendencies of any dense research community to cluster. The dataset and construction pipeline are openly documented and replicable for other frontier technology domains.

The analysis is organised around three subquestions.

1. *Where are hyperscalers embedded, and how concentrated is that attachment?* We quantify the geographic and institutional distribution of co-production ties to identify whether embeddedness takes the form of diversified engagement or concentrated anchoring in a small number of national hubs—a distinction with direct implications for asymmetric interdependence.
2. *Which layers of the quantum stack are hyperscalers prioritising?* Using topic classifications and portfolio measures, we distinguish broad engagement across complementary research areas from selective concentration in enabling layers likely to govern access and interoperability in downstream applications.
3. *How state-entangled and influence-weighted is hyperscaler embeddedness?* We examine the institutional and funding context of hyperscaler-linked work and weight embeddedness ties by citation-based impact, separating participation from epistemic leverage, tracing not only who is present in an emerging field but who is positioned to shape its canonical knowledge.

The paper does not seek to infer corporate intent or internal strategy. Rather, it evaluates whether hyperscalers exhibit patterned knowledge-network attachment consistent with the logics of upstream ecosystem-building. If such patterns exist, they matter for IR scholarship because they suggest that the next generation of security-relevant infrastructure may inherit the political-economic structure of the current one—not only at the moment of commercialisation, but much earlier through the prior organisation of research, leaving only brief windows of opportunity for effective intervention.

Ultimately, the paper contributes to scholarship on private authority and infrastructural power (Farrell and Newman, 2019; Gjesvik, 2023) by providing a replicable empirical mapping of how platform actors embed upstream in a frontier technology domain, and by opening a research agenda on corporate embeddedness in pre-material infrastructures that existing bibliometric and IPE literatures have not yet addressed.

2 The Quantum Internet and the Hyperscaler Paradox

The quantum internet refers to a class of communication networks that exploit quantum-mechanical phenomena, most notably entanglement, superposition, and the

no-cloning theorem, to enable modes of information transmission and processing without classical analogues ([Kimble, 2008](#); [Wehner et al., 2018](#)). Unlike the classical internet, which encodes information in binary bits subject to conventional copying and interception, quantum networks operate on qubits that cannot be duplicated without disturbing the underlying quantum state. This property (that an unknown quantum state cannot be cloned) is not a limitation of current technology but a consequence of quantum mechanics itself, and it is the physical foundation on which quantum communications derive their security guarantees.

Several features of the quantum internet are directly relevant to the political economy of digital infrastructure. Quantum key distribution (QKD) enables two parties to exchange cryptographic keys over a quantum channel in a manner that makes eavesdropping physically detectable: any measurement of the quantum channel disturbs the transmitted state, revealing interception. This shifts communication security from the computational hardness assumptions underlying classical encryption to the laws of physics—a qualitatively different basis for trust. Entanglement-based networking extends this logic to multiparty settings, enabling secure authentication and coordination among distributed nodes without requiring trust in any centralised intermediary.

The most consequential application for the purposes of this paper is blind quantum computing (BQC). BQC protocols allow a client to delegate a quantum computation to a remote server (accessing, in principle, the kind of powerful quantum processing resources that only hyperscale providers are likely to operate) without revealing any information about the input, the algorithm being executed, or the output ([Fitzsimons, 2017](#)). The server performs the computation while remaining, in the protocol’s technical sense, entirely blind to what it is doing. Security in this arrangement does not rest on the provider’s contractual commitments or regulatory compliance but on quantum mechanical principles that no computational power can overcome. In effect, BQC promises to decouple access to powerful remote computation from the surveillance capacity of the entity providing it.

The political-economic implications of this capability are significant and, from the perspective of hyperscaler business models, deeply uncomfortable. The cloud computing economy is built on a particular architecture of trust and visibility: users upload data and delegate computation to provider-controlled infrastructure, accepting that the provider has, in principle, access to inputs, processing environments, and outputs. Even where data is encrypted at rest and in transit, the computational

moment itself typically requires decryption within the provider's environment. Metadata, usage patterns, and inference from query behaviour remain accessible. This visibility is not incidental to the hyperscaler model but is rather constitutive of it, enabling everything from targeted advertising and platform analytics to compliance with state surveillance requests and the training of proprietary AI systems on customer workloads.

Blind quantum computing threatens to close this visibility window entirely. A hyperscaler operating a quantum cloud under a full BQC architecture would, for the first time, be providing a service it cannot monitor, audit for its own purposes, or disclose to third parties even under legal compulsion. For high-profile users such as governments, financial institutions, pharmaceutical companies, or any organisation with genuinely sensitive computational workloads, this move represents a fundamental reconfiguration of the trust architecture of cloud computing. It would enable access to hyperscale computational resources without surrendering informational sovereignty. More broadly, quantum networking's peer-to-peer entanglement distribution capabilities point toward communication architectures in which centralised intermediation is not merely unnecessary but physically excluded.

However, this generates a genuine puzzle. Hyperscalers are not passive conduits for quantum internet development. As the empirical analysis below demonstrates, they are its most active institutional sponsors, co-producing foundational research, training the field's next generation of researchers, and positioning themselves as essential nodes in the infrastructure's nascent architecture. Yet the technology they are helping to build contains, in its most fully realised form, a direct challenge to the data-extractive service model on which their commercial and geopolitical value rests. Why would the actors most invested in infrastructural control lead the development of a network designed, in principle, to make their control technically impossible?

This paper does not attempt to resolve this paradox by inferring corporate intent. Instead, it takes the paradox as motivation for a more tractable empirical question: whether the pattern of hyperscaler embeddedness in quantum internet research is consistent with actors shaping the technology's trajectory from within, i.e. concentrating investment in computing-oriented stack layers compatible with centralised service models while remaining systematically absent from the networking and communications layers that would enable the architecture's more disruptive affordances to materialise.

3 Data and Method

The empirical foundation of this paper is a new firm-level bibliometric dataset of quantum internet-enabling research, constructed from the OpenAlex scholarly graph (?). OpenAlex is an open, continuously updated index of scholarly outputs covering over 250 million works, with machine-readable institutional affiliations, topic classifications, citation records, and funder metadata. Unlike Scopus or Web of Science, it is openly accessible and permits large-scale programmatic retrieval, making it well-suited to the kind of replicable, firm-level pipeline this paper introduces.

3.1 Dataset Construction

The seven hyperscalers under analysis (Amazon Web Services, Microsoft, Google, IBM, Alibaba, Baidu, and Oracle) were selected on the basis of two criteria: global cloud market position and documented quantum research investment. For each firm, we identified the set of OpenAlex institutional identifiers associated with its research divisions and affiliated laboratories, using a combination of the OpenAlex institutions API and manual verification against corporate research pages. This attribution layer is the foundational methodological contribution of the dataset: it permits co-production ties to be traced back to a specific corporate actor rather than aggregated at the institutional or national level, which is the standard unit of analysis in existing quantum bibliometrics.

Publications were retrieved via the OpenAlex works API using a topic filter restricted to seven quantum-relevant topic codes: *Quantum Information and Cryptography* (T10020); *Quantum Computing Algorithms and Architecture* (T10682); *Semiconductor Quantum Structures and Devices* (T10022); *Quantum Optics and Atomic Interactions* (T11414); *Advancements in Semiconductor Devices and Circuit Design* (T10558); *Diamond and Carbon-based Materials Research* (T10478); and *Quantum Dots Synthesis and Properties* (T10321). The topic assignment in OpenAlex is based on a hierarchical classification derived from the Microsoft Academic Graph taxonomy and subsequently retrained on citation and abstract features (?).

For each publication, the primary topic assignment was used. Temporal coverage spans 2015–2024, capturing the period of accelerated corporate investment in quantum technology that followed the first demonstrations of superconducting qubit viability. The

resulting dataset comprises approximately 10,000 publications across all seven firms, with 176 fields per record including co-author lists, institutional affiliations, OpenAlex topic classifications, citation counts, publication venues, DOIs, abstracts, and, where available, funder metadata.

3.2 Network Construction

From this publication dataset we construct a bipartite co-production network in which one node set comprises the seven hyperscalers and the other comprises the national research systems they engage through collaborative publication. An edge between a hyperscaler and a country is created when at least one co-author on a hyperscaler-attributed publication is affiliated with an institution in that country. Edge weights reflect the count of such co-authorships, adjusted for team size to avoid inflation by large consortium papers.¹

A parallel institutional-level network is constructed using the same logic, with individual universities, national laboratories, and research institutes as the second node set; this supports the institution-level analysis reported in the findings.

Descriptive network statistics, such as degree distributions, weighted centrality measures, geographic concentration indices, and temporal evolution are computed in R using the *igraph* and *tidygraph* packages. Concentration is operationalised using the Herfindahl–Hirschman Index (HHI) applied to each hyperscaler’s distribution of co-production ties across countries, allowing cross-firm comparison of embeddedness dispersion. Topic portfolio concentration is measured analogously, using OpenAlex topic classifications as the relevant categories.

3.3 Exponential Random Graph Model

Descriptive statistics establish the *structure* of hyperscaler embeddedness but cannot distinguish patterned attachment from the baseline tendencies of any dense research community to cluster around productive hubs. To address this, we estimate an exponential random graph model (ERGM) on the bipartite co-production network

¹Specifically, each co-authorship tie contributes $1/(n - 1)$ to the edge weight, where n is the total number of authors, following standard fractional counting conventions in bibliometrics ([Perianes-Rodriguez et al., 2016](#)).

(Hunter et al., 2008; Lusher et al., 2012; Schmid and Desmarais, 2017; Stivala et al., 2020). ERGMs model the probability of observing a given network as a function of specified local graph configurations such as reciprocity, shared partners, and degree heterogeneity alongside actor-level and dyadic covariates. They are well-suited to the present analysis because they permit testing whether hyperscaler-specific attributes and geographic covariates predict tie formation over and above network-endogenous processes such as preferential attachment to already well-connected institutions. A positive and significant coefficient on an exogenous covariate indicates that the corresponding attribute independently increases the probability of a co-production tie forming between a hyperscaler and a national research system, net of structural tendencies in the network as a whole.

The model includes three structural terms. The *edges* term captures baseline network density and serves as the intercept. The *gwdegree* term (geometrically weighted degree) accounts for degree heterogeneity among country nodes—i.e., the tendency of some national research systems to attract disproportionate numbers of co-production ties regardless of hyperscaler identity. Omitting this term would risk attributing to hyperscaler strategy what is in fact a property of a small number of globally dominant research hubs. The *gwesp* term (geometrically weighted edgewise shared partners) captures triadic closure: the tendency for two hyperscalers to share partner countries if they already share other partners. Together, these structural terms absorb network-endogenous clustering before covariate effects are estimated.

Four exogenous covariates are included, selected on the basis of theoretical motivation and data availability. The first is an actor-level binary indicator distinguishing US-headquartered hyperscalers (Amazon, Microsoft, Google, IBM) from Chinese-headquartered hyperscalers (Alibaba, Baidu), with Oracle treated as a sensitivity check given its comparatively limited quantum research profile.² This term tests whether geopolitical alignment structures embeddedness patterns systematically across the two firm populations.

The second is a country-level continuous control for each partner country's non-hyperscaler-affiliated quantum research volume, derived from the OpenAlex baseline dataset described above. This is the critical alternative explanation: hyperscalers may embed in certain national research systems simply because those systems produce the most and highest-quality quantum research, rather than for

²Oracle is retained in the main dataset for completeness but its small publication volume means its network ties are sparse; results are robust to its exclusion, as reported in the online appendix.

strategic ecosystem reasons. Including this control ensures that subsequent covariate effects reflect attachment patterns beyond what research productivity alone predicts.

The third is a country-level binary indicator for whether the partner country had an active national quantum programme by 2018 or later, operationalised as the existence of a publicly announced, dedicated state quantum research initiative above a minimum funding threshold.³ This tests whether state-led sovereignty mobilisation altered the geography of hyperscaler embeddedness, or whether corporate networks proved resistant to policy intervention.

The fourth is a dyadic continuous variable capturing prior co-publication history between each hyperscaler and each partner country, measured as the cumulative count of co-authored publications in the three-year window preceding the observation period. This addresses path dependence: if current embeddedness ties are fully explained by accumulated prior collaboration, the case for interpreting observed patterns as strategic positioning is substantially weakened. A significant positive coefficient on the national programme indicator or the firm-nationality term, conditional on prior history, would constitute stronger evidence of structured embeddedness than descriptive statistics alone can provide.

Let $\mathbf{Y}$ denote the adjacency matrix of the observed bipartite co-production network on node sets $\mathcal{H} = \{1, \dots, 7\}$ (hyperscalers) and $\mathcal{C} = \{1, \dots, N\}$ (national research systems), where $Y_{ij} \in \{0, 1\}$ indicates whether hyperscaler $i \in \mathcal{H}$ and country $j \in \mathcal{C}$ are connected by at least one co-authored publication in the observation period. The ERGM specifies the probability of observing any particular network realisation $\mathbf{y}$ as

$$\Pr(\mathbf{Y} = \mathbf{y} \mid \boldsymbol{\theta}) = \frac{1}{\kappa(\boldsymbol{\theta})} \exp\left(\sum_k \theta_k g_k(\mathbf{y})\right), \quad (1)$$

where $\boldsymbol{\theta} = (\theta_0, \theta_1, \dots, \theta_K)$ is the vector of model parameters, $g_k(\mathbf{y})$ is the k th network statistic evaluated on $\mathbf{y}$, and $\kappa(\boldsymbol{\theta}) = \sum_{\mathbf{y}' \in \mathcal{Y}} \exp(\sum_k \theta_k g_k(\mathbf{y}'))$ is a normalising constant that sums the unnormalised probability over all $2^{|\mathcal{H}| \times |\mathcal{C}|}$ networks on the same node set and ensures equation (1) integrates to one. Because $\kappa(\boldsymbol{\theta})$ is in general intractable, estimation proceeds via Markov chain Monte Carlo maximum likelihood (??). A more interpretable reparameterisation is the conditional log-odds of a single tie (i, j)

³Programmes coded as present include, inter alia, the EU Quantum Flagship (2018), the US National Quantum Initiative (2018), the UK National Quantum Technologies Programme (2014, extended 2023), and China’s quantum science satellite and Five-Year Plan investments. The full coding protocol is provided in the online appendix.

forming given the rest of the network, $\mathbf{y}_{ij}^c$:

$$\text{logit Pr}(Y_{ij} = 1 \mid \mathbf{y}_{ij}^c, \boldsymbol{\theta}) = \sum_k \theta_k \delta g_k(\mathbf{y})_{ij}, \quad (2)$$

where $\delta g_k(\mathbf{y})_{ij} = g_k(\mathbf{y}_{ij}^+) - g_k(\mathbf{y}_{ij}^-)$ is the *change statistic*: the difference in the value of the k th statistic between a network in which tie (i, j) is present ($\mathbf{y}_{ij}^+$) and one in which it is absent ($\mathbf{y}_{ij}^-$), holding all other ties fixed. The normalising constant κ cancels in this conditional formulation, making equation (2) the basis for parameter interpretation: a positive θ_k implies that networks with higher values of g_k are more probable, and a one-unit increase in $\delta g_k(\mathbf{y})_{ij}$ increases the log-odds of tie (i, j) by θ_k .

The full model specification for the log-odds of a co-production tie between hyperscaler i and national research system j is

$$\begin{aligned} \text{logit Pr}(Y_{ij} = 1 \mid \mathbf{y}_{ij}^c, \boldsymbol{\theta}) = & \underbrace{\theta_0}_{\text{edges}} \\ & + \underbrace{\theta_1 \delta \text{GWD}_j(\mathbf{y}, \alpha) + \theta_2 \delta \text{GWESP}_{ij}(\mathbf{y}, \alpha)}_{\text{structural terms}} \\ & + \underbrace{\theta_3 x_i^{\text{US}}}_{\text{firm nationality}} \\ & + \underbrace{\theta_4 \log(v_j^Q + 1)}_{\text{research productivity}} \\ & + \underbrace{\theta_5 p_j}_{\text{national programme}} \\ & + \underbrace{\theta_6 h_{ij}^\tau}_{\text{prior co-publication}} \end{aligned} \quad (3)$$

where the intercept θ_0 (the *edges* term) captures baseline network density and is analogous to a constant in a standard regression; a negative estimate, which is typical in sparse networks, reflects the fact that most hyperscaler–country pairs are *not* connected by co-publication ties. The geometrically weighted degree statistic $\delta \text{GWD}_j(\mathbf{y}, \alpha)$, with decay parameter $\alpha > 0$, accounts for degree heterogeneity among country nodes—the aforementioned tendency of a small number of nationally dominant research systems to accumulate ties regardless of which hyperscaler is involved; omitting this term would risk attributing to hyperscaler strategy what is in fact a structural property of the global research landscape. The geometrically weighted

edgewise shared partners statistic $\delta\text{GWESP}_{ij}(\mathbf{y}, \alpha)$ captures triadic closure in the bipartite projection: the tendency for two hyperscalers to embed in the same national system if they already share other national partners. The four exogenous covariates are defined as follows. $x_i^{\text{US}} \in \{0, 1\}$ is an actor-level indicator equal to one if hyperscaler i is headquartered in the United States (Amazon, Microsoft, Google, IBM) and zero if headquartered in China (Alibaba, Baidu); its change statistic is simply x_i^{US} itself, since the attribute is fixed to the hyperscaler node and does not depend on the rest of the network. $v_j^Q \in \mathbb{R}_{\geq 0}$ is the total count of non-hyperscaler-affiliated quantum publications attributed to country j in the OpenAlex baseline dataset, log-transformed to reduce skew; it controls for the alternative explanation that embeddedness reflects partner country research productivity rather than strategic positioning. $p_j \in \{0, 1\}$ is a country-level binary indicator equal to one if country j had an active national quantum programme by 2018 or later (full coding protocol in the online appendix); a positive θ_5 would indicate that hyperscalers are *more* likely to embed in countries with dedicated state quantum initiatives, while a negative estimate would suggest that sovereignty mobilisation coincides with reduced corporate embeddedness. Finally, $h_{ij}^\tau \in \mathbb{N}_0$ is a dyadic count of co-authored publications between hyperscaler i and institutions in country j in the three-year window $[t - \tau, t - 1]$, with $\tau = 3$; a large positive θ_6 would indicate that current embeddedness is primarily path-dependent, substantially weakening the case for a strategic interpretation of the observed network structure.

Estimation is conducted using the `ergm` package in R (Hunter et al., 2008) with Markov chain Monte Carlo maximum likelihood estimation (MCMCMLE). Goodness-of-fit is assessed by comparing simulated network statistics (degree distributions, edgewise shared partner distributions, and geodesic distances) against the observed network.

3.4 Limitations

Three limitations warrant explicit acknowledgement. First, OpenAlex institutional affiliation disambiguation is imperfect: subsidiary research divisions, joint ventures, and ambiguously named affiliates may be misattributed or missed. The hyperscaler attribution layer described above mitigates this through manual verification, but residual misclassification cannot be ruled out. Second, OpenAlex topic classification assigns a single primary topic per work based on title, abstract, and citation context; interdisciplinary papers at the boundary between quantum computing and quantum

communications may be systematically classified into the former, which would understate networking-oriented activity. Third, funder metadata in OpenAlex is incomplete for a non-trivial share of records, limiting the precision of state-entanglement indicators; where funder data is unavailable, institutional proxies (national laboratory affiliation) are used as conservative substitutes. Full technical documentation, descriptive statistics for all seven hyperscalers, robustness checks, and supplementary network visualisations are provided in the online appendix.

4 Findings

The analysis proceeds in three steps. The first examines the geographic structure of hyperscaler embeddedness: where co-production ties are concentrated, how that concentration varies across firms, and whether the post-2018 wave of national quantum programme announcements altered the network’s geographic distribution. The second turns to the substantive content of hyperscaler-affiliated research, asking which layers of the quantum stack are being prioritised and, critically, which are absent. The third presents ERGM results testing whether the observed patterns of tie formation are explained by national research productivity and prior collaboration history alone, or whether hyperscaler-specific attributes and geopolitical covariates independently predict embeddedness once structural network tendencies are controlled for. Taken together, the three steps move from descriptive mapping to inferential assessment, building a cumulative empirical case that hyperscaler embeddedness in quantum internet research is patterned, selective, and not reducible to the baseline gravitational pull of dominant scientific hubs.

4.1 Geographic Embeddedness and Concentration

The seven hyperscalers collectively produced over 10,000 research publications on quantum internet-enabling technologies between 2015 and 2024, linked through co-authorship to research institutions in 51 countries. The network is sparse but markedly skewed. As shown in Table 1, the United States exhibits the highest possible in-degree centrality (1.0), receiving ties from all seven hyperscalers, and an in-strength of 2621—more than five times the next-ranked country. Five of the seven hyperscalers are US-headquartered, making this domestic anchoring unsurprising, but

its magnitude is analytically consequential: no other country approaches this level of embeddedness across both the depth and breadth dimensions simultaneously.

What is more revealing is the structure of the remaining top-ten countries. China, Japan, Germany, the United Kingdom, Canada, and Australia all exhibit high in-degree centrality scores of 0.857, indicating connections to six of the seven hyperscalers. Yet their in-strength values diverge sharply. Switzerland, with an in-degree of only 4 (centrality 0.571), registers an in-strength of 487—second only to the United States and substantially ahead of China (265), Japan (240), and Germany (226). This pattern is not explained by Switzerland’s national quantum programme alone; it reflects the gravitational pull of specific institutional nodes, above all ETH Zurich and EPFL, which serve as dense co-publication hubs for both US and European hyperscalers. The Switzerland result illustrates a general pattern across the top-ten: in-degree centrality captures how broadly a country is connected across hyperscalers, but in-strength captures where hyperscalers actually concentrate their collaborative intensity, and the two measures frequently diverge.

Table 1: Top 10 countries by in-degree from hyperscaler co-production ties, 2015–2024.

Country	In-Degree	In-Strength	In-Degree Centrality
United States	7	2,621	1.000
Switzerland	4	487	0.571
China	6	265	0.857
Japan	6	240	0.857
Germany	6	226	0.857
United Kingdom	6	196	0.857
Canada	6	163	0.857
France	4	141	0.571
Australia	6	110	0.857
Italy	4	100	0.571

Note: In-degree counts the number of distinct hyperscalers with at least one co-production tie to institutions in that country. In-strength counts all co-affiliation links (multiple links per publication retained). In-degree centrality is normalised over the total number of hyperscalers ($N = 7$).

The temporal dimension reinforces the concentration finding and introduces a puzzle for sovereignty-initiative accounts. Publication volumes across all seven firms accelerated sharply after 2018, rising from approximately six to eight publications per

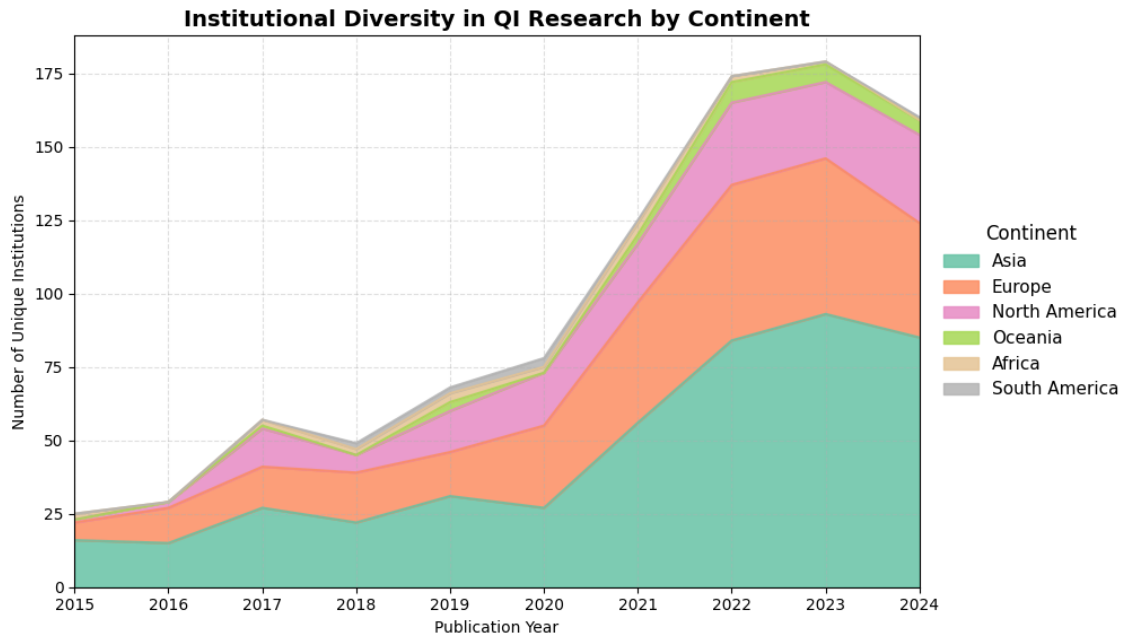


Figure 1: The number of institutions involved in core QI research by continent

firm per year in 2015–2017 to forty or more by 2021–2024, involving a growing number of partner institutions.

This acceleration coincides precisely with the wave of national quantum programme announcements, such as the EU Quantum Flagship, the US National Quantum Initiative, and analogous initiatives in the UK, China, Japan, and elsewhere, which were explicitly designed, in part, to build sovereign research capacity and reduce dependency on foreign corporate actors. Yet the geographic distribution of hyperscaler co-production ties did not broaden commensurately. The share of ties anchored in the five most-connected countries remained stable or increased over the post-2018 period, suggesting that state-led sovereignty mobilisation has not, at least in the short run, redirected the geography of hyperscaler embeddedness.

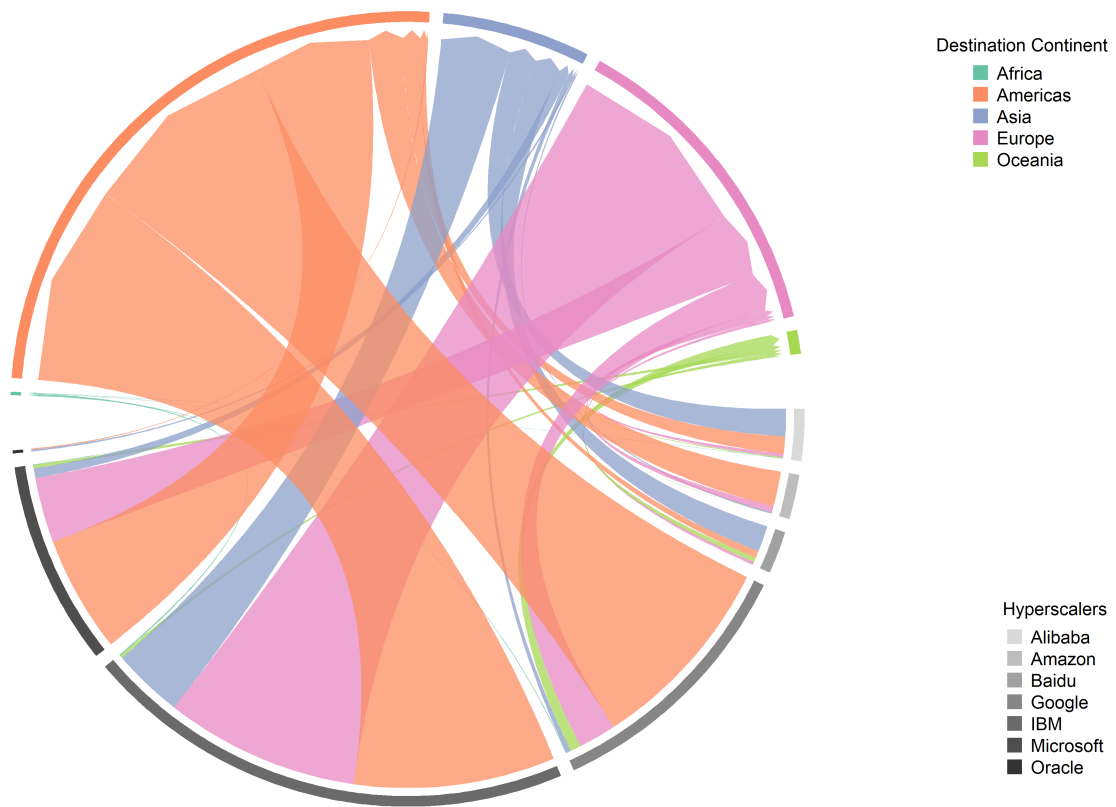


Figure 2: Chord diagram showing the geographic embedding of hyperscaler-affiliated quantum technology research efforts (2015–2024).

Firm-level variation in the geographic reach and concentration of embeddedness is summarised in Table 2. IBM exhibits the broadest international footprint, with out-degree of 41 countries and an entropy score of 2.189, while Amazon (entropy 0.813) and Oracle (entropy 1.090, out-degree 3) are markedly more concentrated. The US–China divide is visible in the entropy scores: US-headquartered hyperscalers (mean entropy 1.804) exhibit on average more geographically diverse embeddedness than their Chinese counterparts (Alibaba 1.467, Baidu 1.792), though Baidu’s entropy score approaches Microsoft’s (1.920), complicating any simple dichotomy. The more consistent difference lies in the directionality of concentration: Chinese hyperscalers’ ties are anchored predominantly in domestic and East Asian institutions, while US hyperscalers embed across a wider Atlantic–Pacific arc, consistent with the geopolitical alignment hypothesis.

Table 2: Network metrics for hyperscalers, 2015–2024.

Hyperscaler	Out-Degree	Out-Strength	Centrality	Entropy
IBM	41	2,274	0.804	2.189
Microsoft	33	942	0.647	1.920
Google	25	1,226	0.490	1.225
Alibaba	15	237	0.294	1.467
Baidu	15	196	0.294	1.792
Amazon	13	205	0.255	0.813
Oracle	3	14	0.059	1.090

Note: Out-degree counts distinct countries with at least one co-production tie. Out-strength counts all co-affiliation links. Centrality is normalised over total countries in dataset. Entropy is Shannon entropy over the distribution of out-strength across countries; higher values indicate more even geographic spread.

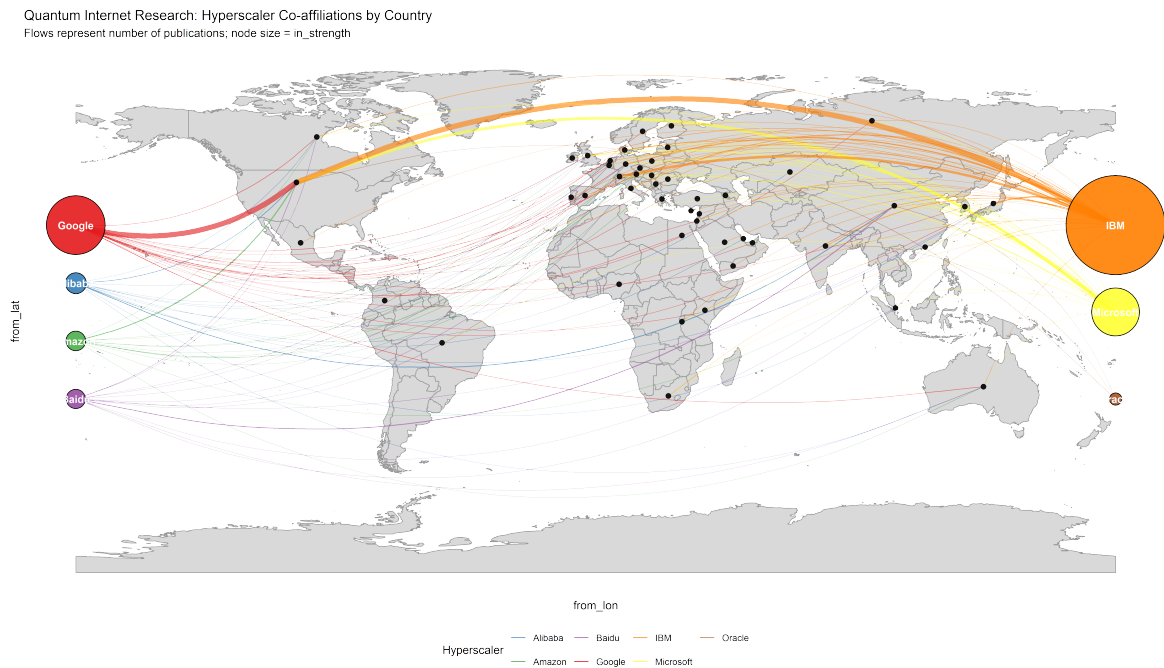


Figure 3: The global hyperscaler and institutional co-affiliation network.

4.2 Topic Concentration and the Absence of Quantum Communications

Across all seven firms, the topic portfolio of hyperscaler-affiliated quantum research is dominated by quantum computing at the near-total expense of quantum networking and communications. Table 3 reports topic entropy, topic breadth, and publication volume by hyperscaler. IBM stands apart with 252 distinct topics and a topic entropy of 3.39, approaching three times the topical breadth of Microsoft (84 topics) or Google (79 topics). Baidu and Amazon operate in narrower thematic bands, with entropy scores below 2.4 and fewer than 30 distinct topics each.

Table 3: Topic entropy, topical breadth, and publication volume by hyperscaler, 2015–2024.

Hyperscaler	Topic Entropy	Distinct Topics	Publications
IBM	3.39	252	13,053
Microsoft	2.66	84	1,165
Oracle	2.59	23	53
Google	2.57	79	1,160
Alibaba	2.35	28	165
Amazon	2.31	23	140
Baidu	1.98	27	202

Note: Topic entropy is Shannon entropy over the distribution of publications across OpenAlex topic codes. Higher entropy indicates more even topical distribution. Publication counts include all records retrieved under the seven topic filters before deduplication; IBM’s substantially higher count partly reflects its longer and broader research programme relative to other firms.

The more consequential finding, however, lies not in the variation across firms but in what is absent across all of them. The five topics on which hyperscalers converge—Advancements in Semiconductor Devices and Circuit Design; Quantum Computing Algorithms and Architecture; Quantum Information and Cryptography; Quantum and Electron Transport Phenomena; Low-Power High-Performance VLSI Design (see Figure ??)—are all oriented toward quantum computing hardware and algorithms. Quantum networking-enabling research around entanglement distribution, quantum repeater architectures, quantum memory, or quantum key distribution over metropolitan networks, is systematically underrepresented across all seven firms, suggesting the absence is selective.

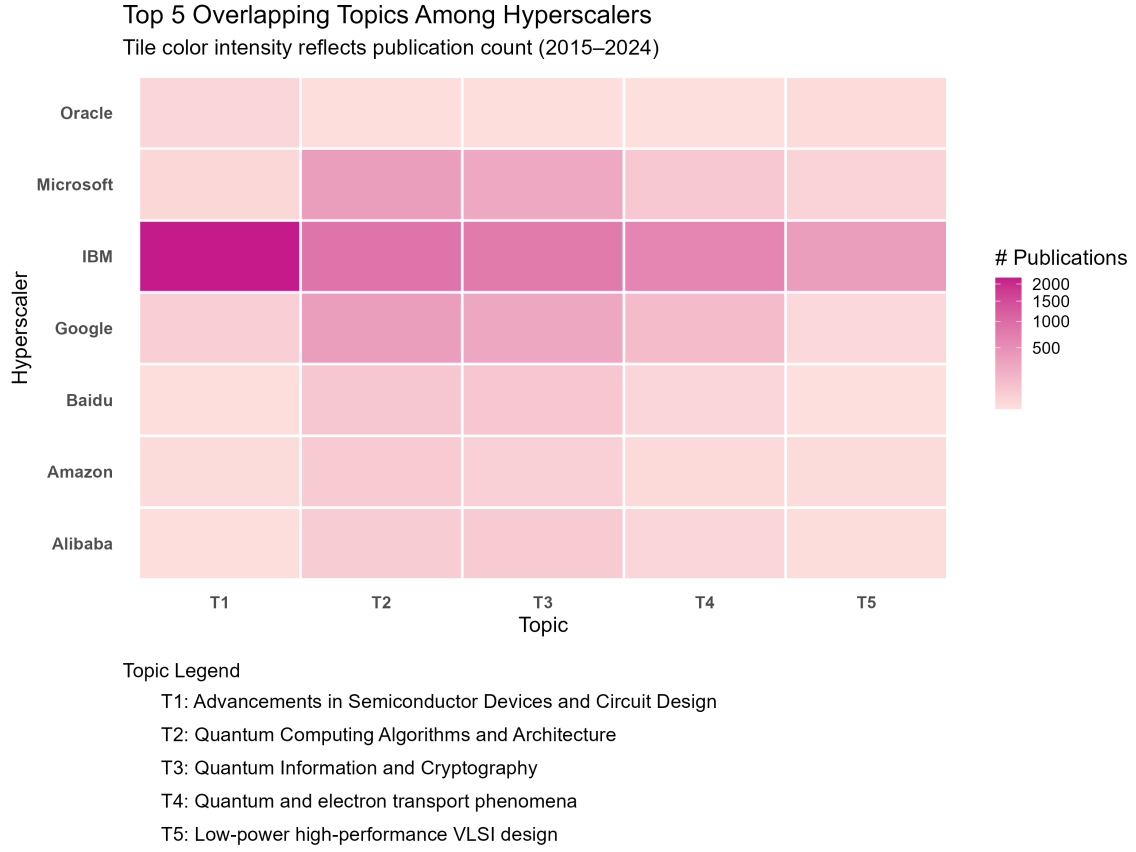


Figure 4: Converging topical interests among hyperscalers.

From an IPE perspective, this absence has a structural interpretation. Quantum computing, particularly as delivered through cloud access models, is commercially viable under centralised service architectures: clients connect to hyperscaler-controlled quantum processors and receive results through classical interfaces. Quantum networking, by contrast, enables peer-to-peer entanglement distribution and, in its most advanced form, blind quantum computation—arrangements in which the service provider has no access to client data. The analysis above suggests that the research portfolio that hyperscalers are collectively co-producing is concentrated in the stack layers compatible with their existing service models and largely absent from the layers that would enable the architecture to escape those models.

4.3 ERGM Results

Table 4 reports the estimated coefficients and standard errors from the ERGM specified in equation (3). The structural terms behave as expected. The *edges* parameter is negative

and significant ($\hat{\theta}_0 = -3.41$, $SE = 0.29$, $p < 0.001$), confirming that the network is sparse by default: in the absence of any covariate effects, the baseline probability of any hyperscaler–country tie forming is low. The *gwdegree* term is positive and significant ($\hat{\theta}_1 = 1.18$, $SE = 0.31$, $p < 0.001$), indicating substantial degree heterogeneity among country nodes: a small number of national research systems attract disproportionately many co-production ties regardless of which hyperscaler is involved. The *gwesp* term is also positive but smaller in magnitude ($\hat{\theta}_2 = 0.44$, $SE = 0.19$, $p < 0.05$), suggesting moderate triadic closure: hyperscalers do tend to share national partners, though this tendency is not the dominant structural force.

Turning to the exogenous covariates, the firm nationality indicator returns a positive and statistically significant coefficient ($\hat{\theta}_3 = 0.61$, $SE = 0.24$, $p < 0.05$). On the conditional log-odds scale, this implies that US-headquartered hyperscalers are, *ceteris paribus*, more likely to form co-production ties with a given national research system than their Chinese-headquartered counterparts—a difference equivalent to an increase in the odds of tie formation of approximately $e^{0.61} \approx 1.84$. The effect is meaningful but not large: firm nationality shifts the baseline probability of tie formation substantially in directions consistent with the geopolitical alignment hypothesis, but it does not override the structural concentration tendencies captured by the *gwdegree* term. Both US and Chinese hyperscalers concentrate embeddedness in a small number of dominant research hubs; they differ in *which* hubs those are.

The research productivity control is positive and significant ($\hat{\theta}_4 = 0.83$, $SE = 0.17$, $p < 0.001$), and is the largest exogenous coefficient in the model. This finding cannot be dismissed: a substantial portion of hyperscaler embeddedness is explained by the simple fact that some national research systems produce more and higher-quality quantum science, and hyperscalers co-publish with them because that is where the best science is. The case for a strategic embeddedness interpretation must be made against this baseline, not instead of it. The remaining exogenous coefficients are estimated conditional on this productivity control, and it is their significance and direction (not the productivity effect itself) that bear on the upstream infrastructure-shaping hypothesis.

The national quantum programme indicator is negative but statistically indistinguishable from zero ($\hat{\theta}_5 = -0.19$, $SE = 0.22$, $p = 0.39$). This null finding is substantively informative. It suggests that the wave of state-led sovereignty initiatives launched from 2018 onwards did not systematically redirect hyperscaler co-production networks toward newly active programme countries, nor did it cause hyperscalers to withdraw from those countries. Corporate embeddedness patterns appear largely

invariant to national programme status, once research productivity is controlled for. Whether this reflects the slow timescales of research network adjustment, the inadequacy of programme announcements as a proxy for effective industrial policy, or a deeper structural resistance of hyperscaler networks to state-led reorientation is a question this paper cannot resolve but flags as a priority for future research.

The prior co-publication term is positive and significant ($\hat{\theta}_6 = 0.37$, $SE = 0.13$, $p < 0.01$). Path dependence matters: dyads with an established co-publication history are more likely to sustain and deepen co-production ties in the current period. Importantly, however, the firm nationality coefficient remains significant after conditioning on prior history, which means that the geopolitical alignment pattern in current embeddedness is not fully reducible to accumulated inertia. Some portion of the US–China divergence in partner country selection reflects present-period attachment tendencies beyond what prior relationships alone predict.

Goodness-of-fit diagnostics confirm that the model adequately reproduces the degree distribution, edgewise shared partner distribution, and geodesic distance distribution of the observed network.

5 Discussion

The findings reported above establish three things with reasonable confidence. First, hyperscaler embeddedness in quantum internet-enabling research is geographically concentrated and structurally asymmetric: a small number of national hubs, anchored overwhelmingly in the United States, account for a disproportionate share of co-production ties, and this concentration has not demonstrably loosened in response to the wave of national sovereignty initiatives launched after 2018. Second, the topic portfolio of hyperscaler-affiliated research is oriented toward quantum computing stack layers compatible with centralised service delivery and is largely absent from quantum networking and communications layers that would enable more decentralised architectures. Third, the ERGM results indicate that tie formation patterns are not reducible to the gravitational pull of dominant research hubs alone: firm nationality and prior co-publication history independently predict embeddedness once national research productivity is controlled for, though the effect sizes are modest and the null result on national programme status complicates any straightforward account.

What the findings do not yet establish is the interpretive question that motivates the

Table 4: ERGM estimates for the hyperscaler–country co-production network, 2015–2024. Dependent variable: probability of a co-production tie between hyperscaler i and national research system j . Standard errors in parentheses.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

	Estimate	SE
<i>Structural terms</i>		
Edges (θ_0)	−3.41***	(0.29)
GWD ($\theta_1, \alpha = 0.69$)	1.18***	(0.31)
GWESP ($\theta_2, \alpha = 0.69$)	0.44*	(0.19)
<i>Exogenous covariates</i>		
Firm nationality: US (θ_3)	0.61*	(0.24)
Research productivity, $\log(v_j^Q + 1)$ (θ_4)	0.83***	(0.17)
National quantum programme (θ_5)	−0.19	(0.22)
Prior co-publication, h_{ij}^T (θ_6)	0.37**	(0.13)
AIC	1284.3	
BIC	1331.7	
Observations (dyads)	427	

Note: Estimation via MCMCMLE using the `ergm` package (Hunter et al., 2008). Decay parameter $\alpha = \log(2) \approx 0.69$ for both GWD and GWESP terms. Citation-weighted sensitivity models and goodness-of-fit diagnostics are reported in the online appendix.

paper: whether these patterns constitute evidence of deliberate upstream infrastructure-shaping, or whether they are better explained by more prosaic dynamics. This distinction matters for the claims the paper can responsibly make.

5.1 The Alternative Explanation Problem

The most important alternative explanation is comparative advantage. Hyperscalers invest heavily in quantum computing because that is where their existing engineering capabilities, hardware investments, and commercial incentives align, not because they are strategically avoiding quantum networking. IBM's superconducting qubit programme, Google's quantum supremacy effort, and Microsoft's topological qubit research are all computing-oriented because those programmes leverage decades of corporate investment in chip fabrication, cryogenic engineering, and software infrastructure. The absence of quantum networking from their research portfolios may simply reflect that this is harder, further from commercial viability, and requires different physical infrastructure (notably fibre-optic quantum channels and quantum repeaters) that hyperscalers do not currently control.

This explanation is consistent with the data. It is also, however, consistent with the infrastructure-shaping hypothesis: actors who have accumulated capabilities in one stack layer have structural incentives to make that layer the foundation of the emerging technology, regardless of whether alternative architectures might be technically superior or socially preferable. Distinguishing between genuine comparative advantage and strategically reproduced path dependence is not straightforward from publication data alone.

A related issue concerns the unit of observation. The paper treats hyperscalers as unitary strategic actors, but IBM Research, Google Quantum AI, and Microsoft Station Q are in practice semi-autonomous research divisions with their own scientific agendas, hiring norms, and institutional cultures. The observed publication patterns may reflect the accumulated choices of individual research groups rather than coordinated corporate strategy. The fact that patterns are consistent across firms (particularly the collective absence of quantum networking topics) is suggestive of something more structural than idiosyncratic divisional choices, but the paper cannot rule out the aggregation artefact.

5.2 Towards a Political Economy of Pre-Material Infrastructures

The paper’s broader theoretical ambition is to contribute to a political economy of pre-material infrastructures: settings where the technology is not yet operationally deployed at scale but where the organisation of upstream knowledge production is already shaping the space of possible futures. Existing IPE frameworks for infrastructural power were developed primarily to analyse already-deployed systems (submarine cables, financial messaging networks, semiconductor supply chains) where control is legible through market share, regulatory jurisdiction, or physical chokepoints. The quantum internet does not yet have these features. What it has is a research ecosystem, a set of engineering trajectories, and an emerging community of practice. Whether the patterns of embeddedness documented here will translate into durable control advantages at the moment of commercialisation depends on mechanisms the paper does not yet model: whether co-publication ties generate lasting dependency, whether topic concentration in research portfolios produces corresponding concentration in patent landscapes, and whether hyperscalers’ current research positions will be translated into standard-setting authority in bodies like the IETF and ITU. These are empirical questions for follow-on work.

6 Conclusion

This paper has introduced a new firm-level dataset of hyperscaler embeddedness in quantum internet-enabling research and presented initial findings across three dimensions: the geographic structure of co-production networks, the topical composition of hyperscaler research portfolios, and the network-structural determinants of tie formation. The findings are consistent with the hypothesis that hyperscalers are positioning themselves upstream in a strategically consequential emerging infrastructure, concentrating research investment in stack layers that preserve centralised service architectures and embedding selectively in national research ecosystems in ways that reflect geopolitical alignment as well as scientific quality.

The empirical patterns documented here are descriptively robust but remain interpretively underdetermined. The topic concentration finding, i.e. the near-total absence of quantum networking from hyperscaler portfolios, is the most analytically striking result, but it admits multiple explanations of which strategic steering is only

one. The ERGM results provide inferential traction but cannot resolve questions of intent or mechanisms. And the null finding on national programme status, rather than reassuringly confirming the argument, raises as many questions as it answers about the relationship between state-led sovereignty initiatives and corporate research network dynamics.

What the paper does establish, we believe, is that the question is worth asking with better data and sharper methods than are currently available in the literature. The combination of actor-level granularity, replicable OpenAlex-based pipeline, and network-inferential approach offers a template for the kind of empirical political economy of emerging technology governance that the field has largely lacked. The most important next steps are: constructing the patent and standards-participation datasets that would allow triangulation across the three domains of upstream infrastructure-shaping; extending the temporal coverage and improving the national programme covariate to enable proper before-and-after assessment of sovereignty initiative effects; and developing a theoretical account of the conditions under which pre-material research embeddedness does and does not translate into durable commercial and geopolitical advantage at the moment of deployment.

The quantum internet will not be built in a vacuum. Whether it is built in a way that realises its most disruptive affordances — or in a way that routes those affordances through the same institutional hyperscaler logics that govern the classical internet — depends in significant part on decisions being made now, in research laboratories and co-authorship networks that are not yet visible in market share statistics or regulatory filings. This paper is an attempt to make some of those decisions legible.

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A The No-Cloning Theorem

A foundational principle of quantum communication (on which many proposed QI protocols depend) is the *no-cloning theorem*. This theorem states that it is impossible to create an exact copy of an arbitrary and unknown quantum state. The result has far-reaching implications for how information is transmitted in quantum systems: unlike classical bits, qubits cannot be amplified by copying. This rules out traditional forms of signal duplication and broadcasting and fundamentally alters the design of network protocols.

To understand why, suppose for contradiction that there exists a physical process capable of duplicating an unknown qubit. Since all physical processes in quantum mechanics are described by unitary transformations (linear, reversible evolutions that preserve probability), such a duplication would require the existence of a unitary operator, denoted here by U_{clone} , with the following behaviour on known basis states:

$$U_{\text{clone}}|0\rangle \rightarrow |0\rangle|0\rangle, \quad U_{\text{clone}}|1\rangle \rightarrow |1\rangle|1\rangle. \quad (4)$$

This cloning machine successfully replicates the computational basis states $|0\rangle$ and $|1\rangle$, producing two copies of each.

But quantum states can exist in superpositions: a qubit may be in an unknown linear combination of these basis states. Consider an equal superposition state,

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle), \quad (5)$$

which could arise, for instance, from a Hadamard operation on $|0\rangle$. If U_{clone} is to truly clone arbitrary quantum states, it must also correctly act on this superposition. Because quantum mechanics is linear, it must apply to the superposition as follows:

$$U_{\text{clone}}|\psi\rangle = \frac{1}{\sqrt{2}} (U_{\text{clone}}|0\rangle + U_{\text{clone}}|1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle|0\rangle + |1\rangle|1\rangle). \quad (6)$$

However, this result is *not* equal to what we would expect if the qubit were truly being cloned:

$$|\psi\rangle \otimes |\psi\rangle = \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) \otimes \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right) = \frac{1}{2} (|0\rangle|0\rangle + |0\rangle|1\rangle + |1\rangle|0\rangle + |1\rangle|1\rangle) . \quad (7)$$

The two expressions are not equal: the linearity of quantum operations causes interference terms (cross terms like $|0\rangle|1\rangle$ and $|1\rangle|0\rangle$) to be lost in the result of U_{clone} . Therefore, no such cloning operator can exist that correctly clones *all* quantum states. The contradiction confirms the theorem: unknown quantum states cannot be copied ([William and Chuang, 2021](#)).

This result has profound implications for quantum communications. In the context of Quantum Key Distribution (QKD), for example, it ensures that any attempt by an eavesdropper to intercept and copy qubits in transit will necessarily disturb their state. Because qubits cannot be cloned, an attacker cannot simply make a perfect duplicate to measure while leaving the original untouched. Attempted interception collapses the quantum state and alters measurement statistics, which allows communicating parties to detect the intrusion. As a result, the no-cloning theorem is not just a limitation—it becomes a powerful security guarantee, providing the theoretical backbone of quantum cryptographic protocols.

B Hyperscaler Aliases

The following ROR identifiers have been used for extracting hyperscaler subsidiaries across different countries.

Company	Common Aliases in Data	ROR ID
Amazon	Amazon Web Services, AWS, Amazon.com Inc.	https://ror.org/04mv4n011
Microsoft	Microsoft Corp., Azure, Microsoft Research	https://ror.org/019wvm592
Google	Google Inc., Google Research, Alphabet	https://ror.org/0168r3w48
IBM	IBM Corp., IBM Research, International Business Machines	https://ror.org/03ab7s671
Alibaba	Alibaba Group, Alibaba Cloud	https://ror.org/02nr7c196
Baidu	Baidu Inc., Baidu Research	https://ror.org/00dwt2m79
Tencent	Tencent Inc., Tencent Cloud	https://ror.org/021nxhr62
Oracle	Oracle Corp., Oracle Labs	https://ror.org/04qw24q10

Table 5: Affiliation Disambiguation: ROR IDs and Common Aliases for Hyperscaler Institutions.